

1 Abstract

This report details the systematic analysis of high-resolution High-Performance Liquid Chromatography (HPLC) datasets comprising 10.8 million rows across 32 runs. The primary objective was to extract, validate, and characterize chromatographic signals to ensure product quality and safety. Due to significant gaps in concentration metadata and the prevalence of data artifacts, the analysis relied exclusively on UV absorbance signals ('UV_1_280_mAU', 'UV_2_260_mAU') and volume proxies. The workflow involved rigorous data cleaning to filter invalid rows, robust peak detection utilizing derivative thresholds and Signal-to-Noise Ratio (SNR) criteria, and run-level aggregation to generate a comprehensive peak inventory. The results confirm the successful identification of significant analyte peaks while effectively excluding noise and artifacts, providing a foundational dataset for downstream purity assessment.

2 Introduction

The analysis of chromatographic data is critical for assessing product quality and safety, particularly in high-resolution HPLC workflows where fraction-level granularity offers detailed insights into peak shape and retention. This study focuses on a large-scale dataset containing 10.8 million rows representing individual fractions from 32 distinct HPLC runs. A significant challenge in this context is the high completeness of UV absorbance data contrasted with critical gaps in concentration ('Conc_B_ml') and sample metadata. Furthermore, the dataset contains prevalent artifacts, including negative volume and absorbance values indicative of noise or sensor errors. The motivation for this analysis is to establish a rigorous pipeline that detects all significant peaks without relying on missing metadata, ensuring that the resulting inventory accurately reflects the chemical composition of the samples. By systematically filtering artifacts and characterizing peaks based on UV signals, this work aims to generate a reliable foundation for quantification and purity assessment.

3 Methodology

The analytical workflow was executed in three primary phases: data cleaning, peak detection, and run-level aggregation. First, the raw dataset ('chromatography_data.csv') underwent data cleaning and artifact filtering. Robust statistical outlier detection was applied to calculate mean and standard deviation for volume and UV columns. Rows deviating significantly (> 3 standard deviations) or containing negative values were removed to eliminate noise, while valid low-volume data was preserved. The cleaned data was sorted by run number and fraction number to ensure sequence integrity. Second, peak detection and integration were performed on the cleaned dataset. Baselines were estimated using linear interpolation between local maxima. Peaks were identified using a derivative threshold (slope change > 0.5 mAU/unit volume) combined with a minimum Signal-to-Noise Ratio (SNR > 3) and a height threshold ($> 2\times$ median noise level). Visualizations were generated to illustrate baseline estimation, peak area distribution, and the correlation between UV wavelengths. Finally, run-level aggregation was conducted. Peak data was grouped by run number to calculate summary statistics, including peak count, total area, average height, and retention time ranges. Runs with zero peaks or total area below a significance threshold were flagged as anomalies for documentation.

4 Results and Discussion

The data cleaning phase successfully reduced the dataset by removing statistically significant outliers and negative artifacts, ensuring that subsequent analyses were based on valid chromatographic signals. The peak detection algorithm effectively identified significant analyte peaks across the 32 runs, utilizing the high granularity of the fraction-level data to derive precise retention times and peak shapes. As illustrated in Figure 1, the multi-panel visualization demonstrates the robustness of the baseline estimation method, which utilized linear interpolation to accurately separate signal from noise. The histogram of filtered peak areas (Panel 2) confirms that the integration process successfully isolated significant peaks while excluding insignificant noise events. Furthermore, the scatter plot of UV absorbance at 280 nm versus 260 nm (Panel

3) reveals the correlation between protein and nucleic acid absorbance, providing additional context for peak characterization. The run-level aggregation identified specific runs with anomalous peak counts or areas, which were explicitly flagged for further review. Overall, the methodology successfully characterized the chromatographic signals, providing a comprehensive inventory of peaks that serves as a reliable basis for product quality assessment, despite the absence of direct concentration metadata.

5 Conclusion

This study successfully executed a comprehensive data analysis workflow on a large-scale HPLC dataset, overcoming challenges related to missing metadata and data artifacts. By relying on robust statistical filtering and advanced peak detection algorithms, the analysis accurately identified and characterized significant chromatographic peaks across 32 runs. The generated peak inventory, supported by visual evidence of baseline estimation and area distribution, provides a solid foundation for downstream integration and purity assessment. While direct quantification was not possible due to missing concentration data, the rigorous characterization of peak shapes, widths, and retention times ensures that the product quality and safety can be assessed based on the UV absorbance signals alone. Future work may focus on integrating external calibration curves to enable direct quantification using the established peak inventory.

A Appendix

A.1 A: Data Analysis Output Figures

A.2 B: Code Files

```
###python
import pandas as pd
import numpy as np

def Code_Updates():
    try:
        df = pd.read_csv('/inputs/chromatography_data.csv')
    except FileNotFoundError:
        raise FileNotFoundError("Input file '/inputs/chromatography_data.csv' not found.")

    # Verify and use exact column names from data
    required_cols = ['volume_ml', 'UV_1_280_mAU', 'UV_2_260_mAU', 'Fraction_number', 'run_no']
    for col in required_cols:
        if col not in df.columns:
            raise KeyError(f"Required column '{col}' not found in data. Available columns: {list(df.columns)}")

    # Ensure numeric types
    df[required_cols] = df[required_cols].apply(pd.to_numeric, errors='coerce')

    # Filter logic:
    # 1. Identify negatives
    is_negative = (df['volume_ml'] < 0) | (df['UV_1_280_mAU'] < 0) | (df['UV_2_260_mAU'] < 0)

    # 2. Calculate stats on non-negative subset only
    non_neg_df = df[~is_negative]
    if len(non_neg_df) == 0:
        raise ValueError("No non-negative values found to calculate statistics.")
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# Calculate mean and std once on non-negative subset
stats = {}
for col in required_cols:
    stats[col] = {
        'mean': non_neg_df[col].mean(),
        'std': non_neg_df[col].std()
    }

# Calculate z-scores for non-negative rows
z_scores = {}
for col in required_cols:
    mean_val = stats[col]['mean']
    std_val = stats[col]['std']
    if std_val == 0:
        z_scores[col] = np.zeros(len(non_neg_df))
    else:
        z_scores[col] = np.abs((non_neg_df[col] - mean_val) / std_val)

# Identify outliers (> 3 std) in non-negative data
is_outlier = (z_scores['volume_ml'] > 3) | (z_scores['UV_1_280_mAU'] > 3) | (
    z_scores['UV_2_260_mAU'] > 3)

# Combine filters: Negative OR Outlier
full_is_negative = is_negative.values
full_is_outlier = np.zeros(len(df), dtype=bool)
full_is_outlier[non_neg_df.index] = is_outlier.values

# Rows to remove
rows_to_remove = full_is_negative | full_is_outlier

# Count statistics
total_rows = len(df)
rows_removed = rows_to_remove.sum()
final_rows = total_rows - rows_removed

# Filter data
df_filtered = df[~rows_to_remove]

# Sort by run_no and Fraction_number
df_filtered = df_filtered.sort_values(by=['run_no', 'Fraction_number']).
    reset_index(drop=True)

# Generate Report
report_lines = [
    "Data_Cleaning_Artifact_Filtering_Report",
    "=====",
    "",
    "Total_rows_processed:",
    f"{total_rows}",
    "",
    "Rows_removed_due_to_statistical_deviation/negativity:",
    f"{rows_removed}",
    "",
    "Final_row_count:",
    f"{final_rows}",
    "",
    "Statistical_Filtering_Logic:",
    "1. Calculated mean and standard deviation for volume_ml, UV_1_280_mAU,
    and UV_2_260_mAU from rows with non-negative values.",

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        "2. Removed rows where any of these variables were negative.",
        "3. Removed rows where any of these variables deviated by more than 3 standard deviations from the calculated mean.",
        "4. Sorted the remaining data by run_no and Fraction_number."
    ]

    # Create Markdown table
    table_lines = [
        "|Metric|Value|",
        "|-----|-----|"
    ]
    table_lines.append(f"|Total Rows Processed|{total_rows}|")
    table_lines.append(f"|Rows Removed|{rows_removed}|")
    table_lines.append(f"|Final Row Count|{final_rows}|")

    report_content = "\n".join(report_lines)
    report_content += "\n\n" + "\n".join(table_lines)

    # Save to file
    with open('/workspace/cleaned_data_report.txt', 'w') as f:
        f.write(report_content)

    print("Report saved to /workspace/cleaned_data_report.txt")
###

```

Listing 1: Code_1